

Theory of Automata

CS-3005

Lecture 11

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Contents

- Closure Properties
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Closure properties

- A language that can be defined by a regular expression is called a regular language.
- Not all languages are regular, as we shall see in the next lecture.
- In this lecture we will focus on the class of all regular languages and discuss some of their properties.

Theorem

If L_1 and L_2 are regular languages, then $L_1 + L_2$, L_1L_2 , and L_1^* are also regular languages.

Notes:

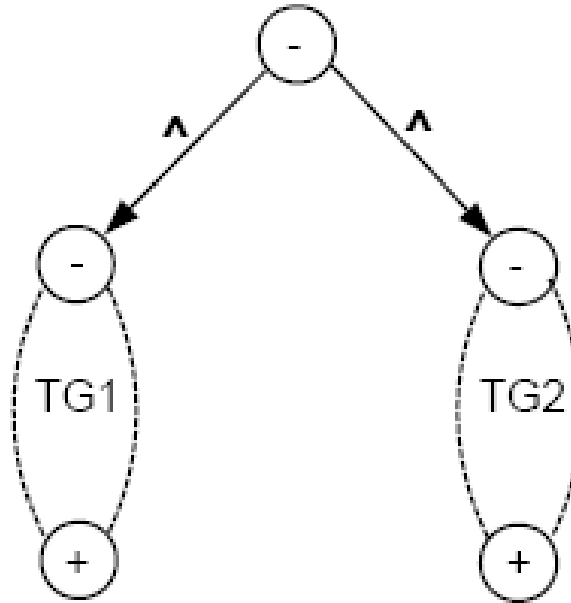
- $L_1 + L_2$ is the language of all words in either L_1 or L_2 .
- L_1L_2 is the product language of all words formed by concatenating a word from L_1 with a word from L_2 .
- L_1^* is the language of all words that are the concatenation of arbitrarily many factors from L_1 .
- The result stated in Theorem 10 is often expressed as “**The set of regular languages is *closed* under union, concatenation, and Kleene closure**”.

Proof by Machine

- Because L_1 and L_2 are regular languages, there must be TGs that accept them (by Kleene's theorem).
- Let TG_1 accepts L_1 and TG_2 accepts L_2 .
- Assume that TG_1 and TG_2 each have a **unique start state** and a **unique separate final state**. If this is not the case originally, then we can modify the TGs so that this becomes true as in Kleene's theorem, Part 2 of the proof (page 93).

Proof

- Then the TG described below accepts the language $L_1 + L_2$.



- By Kleene's theorem, since $L_1 + L_2$ is defined by this TG, it is also defined by a regular expression and hence is a regular language.

Proof

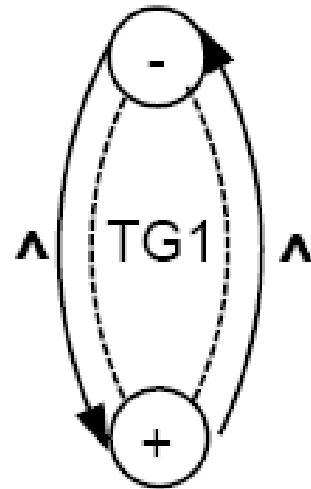
- The TG described below accepts the language L_1L_2 where state 1 is the former + of TG_1 and state 2 is the former - of TG_2 .



- Since L_1L_2 is defined by this TG, it is also defined by a regular expression by Kleene's theorem, and therefore it is a regular language.

Proof

- The TG described below accepts the language L_1^* .



- We begin at the - state and trace a path to the + state of TG1. At this point, we could stop and accept the string or jump back, at no cost, to the - state and run another segment of the input string.

Complements

Definition:

- If L is a language over the alphabet Σ , we define its **complement** L^c to be the language of all strings of letters from Σ that are not words in L .

Example:

- Let L be the language over the alphabet $\Sigma = \{a; b\}$ of all words that have a **double** a in them.
- Then, L^c is the language of all words that do **not** have a double a in them.
- Note that the complement of L^c is L . That is

$$(L^c)^c = L$$

Theorem

If L is a regular language, then L^c is also a regular language. In other words, the set of regular languages is closed under complementation.

Proof

- If L is a regular language, then by Kleene's theorem, there is some FA that accepts the language L .
- Some states of this FA are final states and some are not. Let us reverse the status of each state: **If it was a final state, make it a non-final state.** If it was a non-final state, make it a final state. The start state gets reversed as follows: $- \leftrightarrow \pm$
- If an input **string formerly ended in a non-final state**, it now ends in a final state, and vice versa.
- The new **machine we have just built accepts all input strings that were not accepted by the original FA, and it rejects** all the input strings that used to be accepted by FA.
- Therefore, this **machine accepts exactly the language L^c** . So, by Kleene's theorem, L^c is regular.

Intersection Theorem

If L_1 and L_2 are regular languages, then $L_1 \cap L_2$ is also a regular language. In other words, the set of regular languages is closed under intersection.

Proof

- By DeMorgan's law (for sets of any kind):

$$L_1 \cap L_2 = (L_1^c + L_2^c)^c$$

- This means that the language $L_1 \cap L_2$ consists of all words that are **not** in either L_1^c or L_2^c .
- Because L_1 and L_2 are regular, then so are L_1^c and L_2^c by Theorem 11.
- Since L_1^c and L_2^c are regular, so is $L_1^c + L_2^c$ by Theorem 10.
- Now, since $L_1^c + L_2^c$ is regular, so is $(L_1^c + L_2^c)^c$ by Theorem 11.
- This means $L_1 \cap L_2$ is regular, because

$$L_1 \cap L_2 = (L_1^c + L_2^c)^c \text{ by DeMorgan's law.}$$